Combinatorics
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

1 Donald, Melania and five other people get on a bus one at a time. In how many ways can the seven people board the bus if Melania refuses to get on immediately following Donald? 2

No restrictions: $7!$ orderings

Melania immediately behind Trumpler:

$\boxed{DM}$ _ _ _ _ _ $6!$ orderings

Answer: $7! - 6! = \boxed{4320}$ [2]

2 An exam sat by 130 students was marked out of 25 marks. No half marks were awarded and exactly four students scored full marks. A total of N students received the modal mark. 2

What is the smallest value of N? Justify your answer.

Remove the known results - 4 scores of 25

Categories: the 25 remaining possible marks - 0 to 24

Objects to be categoried: the 126 remaining students $= 5 \times 25 + 1$

As there are more objects than 5 times the number of categories, at least one category must contain 6 objects

ie, at least one mark was awarded to at least six students [2]

3 A 4-digit number begins with 1 and has exactly one pair of identical digits (not necessarily adjacent).

a. How many such numbers can be formed if the repeated digit is 1? 1

Choose position for 2nd '1': 3 ways

Choose 2 different ordered digits from remaining 9 to fill other 2 spots: ${}^9P_2$

$3 \times {}^9P_2 = \boxed{216}$ [1]

b. How many such numbers can be formed in total? 2

Case 2: '1' is not the repeated digit

Choose and place non-repeated digit: 9×3

Choose repeated digit to fill other 2 spots: 8

$9 \times 3 \times 8 = 216$

Total: $\boxed{432}$ [2]

4 From the set of integers 1, 2, 3, … ,999, numbers are selected one at a time without repetition. 2

How many numbers are needed to guarantee that we have chosen at least four multiples of 9?

888 non-multiples + 4 multiples = 892

5 By considering a group of students consisting of n boys and n girls, or otherwise, prove the identity: 3

$$\binom{2n}{3} = 2\binom{n}{3} + 2n\binom{n}{2}$$

LHS = no. ways of choosing 3 students from 2n students = $\binom{2n}{3}$

RHS : 4 cases : Choose 3 boys from n : $\binom{n}{3}$ ways

or Choose 2 boys from n & 1 girl from n : $\binom{n}{2} \cdot \binom{n}{1} = n\binom{n}{2}$

Other 2 cases identical for more girls than boys

$\therefore$ Total ways $= 2\left[\binom{n}{3} + n \cdot \binom{n}{3}\right]$

$= 2\binom{n}{3} + 2n\binom{n}{2}$

As we are counting the same thing on LHS & RHS, result holds.

[3]

6 At a soccer club a team of 13 players is to be chosen from a pool of 25 players consisting of 20 female players and 5 male players. What is the probability that the team will consist of only female players? 2

$$\frac{{}^{20}C_{13}}{{}^{25}C_{13}} = \frac{12}{805}$$

7 How many different ways can 4 people be chosen from a group of 20 people? 1

$${}^{20}C_4 = 4845$$

8 Six chairs are to be painted. Two of the chairs are to be painted black. The other four are to be painted blue, green, yellow and grey. The chairs are then arranged in a row. How many possible arrangements are there for the colours of the chairs? 1

There are 6 chairs but 2 of them are indistinguishable from each other.

The total arrangements can be represented thusly: $\frac{6!}{2!} = 360$

9 Marty the Martian has an infinite number of red, blue, yellow, and black socks in a drawer. If Marty is pulling out socks in the dark, what is the smallest number of socks that Marty must pull out of the drawer to guarantee getting ten socks of the same colour? 1

With 9 red, 9 blue, 9 yellow and 9 black socks

Marty still does not have 10 of a single colour.

Current total = 36

$\therefore$ The 37th sock must mean he meets the required condition.

10 In a class of 27 students, there are 14 boys and 13 girls. The class needs to elect two boys and two girls for the student council.

a. In how many ways could the representatives be chosen? 2

$${}^{14}C_2 \times {}^{13}C_2 = 7098$$

b. If the class elected Henry and Olivia for the student council, in how many ways could the representatives now be chosen? 1

$$13 \times 12 = 156$$

11 Four female and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate? 1

Seat females in $(4-1)!$ ways. Then seat males in $4!$ ways

$\therefore$ number of ways $= 3! \times 4!$

12 A bag contains n red marbles and one blue marble. Three marbles are drawn (without replacement). The probability that the three marbles are red is $\frac{5}{8}$. Find the value of n. 2

$$\text{Prob}(3 \text{ red marbles drawn}) = \frac{{}^{n}C_3}{{}^{n+1}C_3}$$

$$\frac{n}{n+1} \times \frac{n-1}{n} \times \frac{n-2}{n-1} = \frac{5}{8}$$
$$\frac{n-2}{n+1} = \frac{5}{8}$$
$$n = 7$$

$$\frac{{}^{n}C_3}{{}^{n+1}C_3} = \frac{5}{8}$$
$$\frac{\left(\frac{n(n-1)(n-2)}{3\times2\times1}\right)}{\frac{(n+1)(n)(n-1)}{3\times2\times1}} = \frac{5}{8}$$
$$\frac{n-2}{n+1} = \frac{5}{8}$$
$$8n-16 = 5n+5$$
$$n = 7$$

13 Quentin, Elliott and five friends arrange themselves at random in a circle. What is the probability that Quentin and Elliott are not together? 1

Seven people so $(7-1)!$ options.

2 sit together:, 6 blocks of people $2!\,5!$

$P() = \frac{2!\,5!}{6!} = \frac{1}{3}$ so $1 - \frac{1}{3} = \frac{2}{3}$

14 Three cards are chosen from random from a standard pack of 52 cards. What is the probability that one is an Ace, one is a King and one is a Queen? 2

$P(1\times A, 1\times K, 1\times Q) = \frac{\binom{4}{1}\binom{4}{1}\binom{4}{1}}{\binom{52}{3}} = \frac{64}{22\,100} = \frac{16}{5525}$ #

15 A school mathematics department consists of 5 male and 5 female teachers. How many different exam-writing committees comprising three teachers could be formed if there must be at least one teacher of each sex on the committee? 1

2M1F ${}^5C_2 \times {}^5C_1$

2F1M ${}^5C_2 \times {}^5C_1$

At least one of each sex $2 \times {}^5C_2 \times {}^5C_1 = 2 \times \frac{5\times4}{2} \times 5 = 100$

16 In a barrel there are 50 marbles of various colours. Of these, 5 are green, 17 are blue, 12 are yellow, 12 are purple and 4 are orange. What is the least number of marbles that can be selected from the barrel to ensure that 7 of the selected marbles are of the same colour? 2

Worst Case 5G; 6B; 6Y; 6P; 4O has no 7 of any colour. So the next marble of B, Y, P will give 7 of a colour

$\therefore$ 27 + 1 = 28

28 marbles

17 Each of the students in an athletics team is randomly allocated their own locker from a row of 100 lockers. What is the smallest number of students in the team that guarantees that two students are allocated consecutive lockers? 1

50 spaces is max possible with one gap therefore 51

Combinatorics
Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

1 Consider the word STATISTICS.

a. How many arrangements of the letters are there? $\dfrac{10!}{3!\,3!\,2!} = 50\,400$ 1

b. How many arrangements of the letters are there where the A and C are next to each other? 2

Block the A and C together
This can be done 2! ways
So you have $\boxed{AC}$ _ _ _ _ _ _ _ _

$\dfrac{2!\,9!}{3!\,3!\,2!} = 10080$

2 In how many ways can six people from a group of 15 people be chosen and then arranged in a circle? 1

Choose 6 from 15 $= {}^{15}\mathbf{C}_6 = \dfrac{15!}{9! \times 6!}$

Arrangements of 6 in a circle $= 5!$

Total arrangements $= \dfrac{15!}{9! \times 6!} \times 5! = \dfrac{15!}{9! \times 6}$

3 A 3 × 3 grid is to be filled with numbers from the set {−1, 0, 1}. Prove that among the sums by rows, columns and diagonals, there are at least 2 of these sums equal. 2

There are 7 possible sums {−3, −2, −1, 0, 1, 2, 3}
These are the pigeonholes
There are 3 rows, 3 columns and 2 diagonals will result in 8 calculations. These are the pigeons.
∴ By the pigeonhole principle, at least 2 calculations will result in the same sum.

4 12 cards are drawn from a regular deck of 52 playing cards which includes four kings and four queens. It is known that out of these 12 cards, exactly 2 are kings. What is the probability that there will be more queens than kings? 2

$$P = \frac{N(3 \text{ Queens } \& \text{ 2 Kings}) + N(4 \text{ Queens } \& \text{ 2 Kings})}{N(2 \text{ Kings})}$$

$$= \frac{({}^4C_2 \times {}^4C_3 \times {}^{44}C_7) + ({}^4C_2 \times {}^4C_4 \times {}^{44}C_6)}{{}^4C_2 \times {}^{48}C_{10}} = \frac{53}{2168}$$

Choose 2 Kings

Choose 3 Queens

Choose 7 other cards (not K or Q)

Choose 10 other cards, not Kings.

5 A family of eight is seated randomly around a circular table. What is the probability that the two youngest members of the family sit together? 1

Sit youngest person in one seat.
Total combinations around table $= 7!$
Combinations with youngest 2 side by side $= 2!6!$
$\therefore P = \frac{6!2!}{7!}$

6 A set of 20 students is made up of 10 students from Year 11 and 10 students from Year 12. Five students are to be selected from the set of 20. The order of selection is unimportant.

a. What is the total number of possible selections? 1

total number of possible selections: ${}^{20}C_5 = \binom{20}{5} = 15504$ ✓

b. What is the total number of selections, if there are at least two Year 11 students and at least two Year 12 students in the group of five? 2

$\binom{10}{2} \times \binom{10}{3} + \binom{10}{3} \times \binom{10}{2} = 2 \times 5400 = 10800$ ✓

7 Each of the letters A, A, C, H, M, and S is written on a separate card. The cards are drawn at random from a hat and placed next to each other to form a word. What is the probability that the word ASCHAM appears? 1

$P(ASCHAM) = \frac{1}{\left(\frac{6!}{2!}\right)} = \frac{1}{360}$

8 5 boys and 5 girls are seated at a round table. In how many ways can this happen with Robert and Hannah sitting together? 1

$8! \times 2! = 80640$

9 Given that n students are to be placed into 4 classrooms, what is the minimum number of students to guarantee that there are at least two students born in the same month in each of the 4 classrooms? 1

52

Worst case scenario, in each of the 4 classrooms, there are 12 students all born in different months. By the pigeonhole principle, one more student in each classroom would guarantee that there are at least two students in each classroom who are born in the same month. Therefore, a minimum of 52 students is required to guarantee that at least two students in each classroom are born in the same month.

10 A private institution course regulation requires that the same number of students achieve each grade from A to E where possible. What is the smallest number of students required to ensure at least one particular grade is awarded 7 times? 1

There are 5 grades from A to E. At least one grade awarded 7 times implies each group has at least 6.

$5 \times 6 + 1 = 31$

$\therefore$The least possible number of students in this group would be 31 students.

11 Four females and four male students are to be seated around a circular table. In how many ways can this be done if the males and females must alternate? 1

$3! \times 4!$

12 How many numbers greater than 6000 can be formed with the digits 1, 4, 5, 7, 8 if no digit is repeated? 2

4 digit numbers $= 2 \times 4 \times 3 \times 2 = 48$

5 digit numbers $= 5!$

Total $= 168$

13 A company has a board of twelve directors. Six of these were selected at random to be candidates in the election for the positions of President, Vice-President and Treasurer. In how many ways can these three senior positions be filled? 1

There are ${}^{12}C_6$ ways of selecting the 6 directors to go through to the election.

There are ${}^{6}P_3$ ways of electing people to the 3 positions

$${}^{12}C_6 \times {}^{6}P_3 = \frac{12!}{6!6!} \times \frac{6!}{3!} = \frac{12!}{6!\,3!} = \frac{{}^{12}P_6}{3!}$$

14 Use the pigeonhole principle to determine how many integers must be selected from the numbers 5, 6, 7, 8, 9 and 10 so that at least 2 of the numbers will have a difference of 3? 2

Considering the pairs of numbers which have a difference of 3:

5, 8 6, 9 7, 10

If we selected 5, 6, 7 then we would not have any pairs with a difference of 3. But on our next selection, either 8, 9 or 10 we will have one of the pairs above.

By the time you have selected 4 digits, you must have at least one pair.

15 A mathematician wants to put a total of eight different books on her shelf. She can choose from eight different algebra books and eight different calculus books. Given that the book 'Calculus I' must be on the shelf, how many ways can she select he remaining books to go on the shelf if she wants at least four algebra books and at least two other calculus books? 1

Scenario 1:

4 algebra and 3 other calculus books

${}^{8}C_4 \times {}^{7}C_3 = 2450$

Scenario 2:

5 algebra and 2 other calculus books

${}^{8}C_5 \times {}^{7}C_2 = 1176$

Therefore, the total is $2450 + 1176 = 3626$.

16 A total of five players is selected at random from four sporting teams. Each of the teams consists of ten players numbered 1 to 10.

a. What is the probability that of the five selected players, three are numbered "6" and two are numbered "8"? 2

There are $4 \times 10 = 40$ players in the four teams. Of these, 4 players numbered 6 and 4 players are numbered 8:

3 players from the 4 players that are numbered 6 ie $^4C_3 = 4$.

2 players from the 4 players are numbered 8 ie $^4C_2 = 6$.

$\therefore {}^4C_3 \times {}^4C_2 = 24$

No of ways of selecting 5 players from 40 players is $^{40}C_5$

$\therefore$ Reqd prob $= \frac{24}{^{40}C_5} = \frac{1}{27417}$

b. What is the probability that of the five selected players contain at least four players from the same team? 2

No of selection of atleast 4 players from the 10 players in Team A

= Team A 4 + Team B + 1 another team.

5 players from team A.

$= {}^{10}C_4 \times {}^{30}C_1 + {}^{10}C_5 = 6552$

$\therefore$ P(of selecting atleast 4 players from team A)

$= \frac{6552}{^{40}C_5}$

$\therefore$ Total prob: [Team B, C, D]

$= \frac{6552}{^{40}C_5} \times 4 = \frac{28}{703}$

2

16 In how many ways can all of the letters of the word PYTHAGORAS be arranged in a circle so that the consonants stay together? 1

PYTHAGORAS

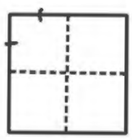
7 consonants, 3 vowels.

"4" items to arrange – 3!

but 7! ways to arrange the consonants.

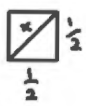
$\frac{3!7!}{2} = \frac{30240}{2} = 15120$

17 Five distinct points are placed randomly within a square. Show that there must exist a pair of points that are at most $\frac{\sqrt{2}}{2}$ distance apart. 2



4 pigeonholes and 5 pigeons. Hence, there must be at least 2 points within one pigeon hole.

Each pigeonhole is a $\frac{1}{2} \times \frac{1}{2}$ unit square.



$x = \sqrt{(\frac{1}{2})^2 + (\frac{1}{2})^2}$

$= \sqrt{\frac{2}{4}} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

Combinatorics
Topic Test 3 Solution

Total Marks: 36

Time: 50 minutes

1 A bag contains 2 red, 5 blue, 6 white, 11 green and 14 yellow marbles. What is the minimum number of marbles that need to be chosen randomly from the bag to ensure that 6 marbles of the same colour have been chosen? 1

$$23 = 2 + 5 + 5 \times 3 + 1$$

2 A class of fifteen mathematics students are seated around a circular table to discuss mathematical problems.

a. In how many different ways can the students be arranged? 1

$$\text{Ways} = (15-1)! = 14!$$

b. If the seats are randomly assigned, find the probability that four particular students, Aidan, Christopher, Daniel and Matthew, are not all seated together as a group of four? Give your answer in the simplest form. 2

Let E be the event where Aidan, Christopher, Daniel and Matthew are seated together.

$$P(E) = \frac{(12-1)! \times 4!}{14!} = \frac{1}{91} \qquad P(\overline{E}) = 1 - \frac{1}{91} = \frac{90}{91}$$

3 There are 96 students in a school who have a birthday in October. What is the minimum number of these students that share the same birthday? 1

$$\frac{96}{31} = 3.09677\ldots \quad \text{i.e. 4 people}$$

4 A Year 12 Biology class consists of 10 girls and 15 boys. In how many ways can a group of three boys and two girls be chosen from this class for a group to work on an investigation project? 1

The girls can be chosen in ${}^{10}\mathbf{C}_2$ ways

The boys can be chosen in ${}^{15}\mathbf{C}_3$ ways

The number of ways the group is chosen = ${}^{10}\mathbf{C}_2 \times {}^{15}\mathbf{C}_3 = 20\ 475$

5 A Mathematics department consists of 12 teachers. Nine of the teachers wear glasses and three of the teachers do not wear glasses. Five of these teachers go out to dinner together. In how many ways will there be more teachers who wear glasses than teachers who do not wear glasses in the group who go out for dinner? 2

5 glasses: $\binom{9}{5}$ 4 glasses, 1 no glasses: $\binom{9}{4} \times \binom{3}{1}$ 3 glasses, 2 no glasses: $\binom{9}{3} \times \binom{3}{2}$

$$\text{total} = \binom{9}{5} + \binom{9}{4} \times \binom{3}{1} + \binom{9}{3} \times \binom{3}{2} = 756$$

6 How many numbers greater than 3 000 can be formed from the digits 1, 3, 5, 7, 9 if no repetition is permitted? 2

1, 3, 5, 7, 9 ; Numbers > 3000

5 digit numbers = 5! = 120

4 digit numbrs = 4 × 4! = 96

Total = 120 + 96 = 216 numbers.

7 What is the number of possible arrangements of the letters in the word LOGARITHMS, if 'G' is next to 'R'? 2

10 letters ~~or~~ GR _ _ _ _ _ _ _ _ _

RG

Number of arrang. = 9! × 2! = 725760

8 In how many ways can a committee of 3 women and 4 men be chosen from 8 women and 7 men if two particular women cannot serve on the committee together? 2

only 1 of the 2 particular woman chosen

$^6C_2 \times 2 \times {}^7C_4 = 1050$

both of the women <u>not</u> chosen

$^6C_3 \times {}^7C_4 = 700$

Total ways = 1750

9 Mr and Mrs Antony (the hosts) and 6 guests sit around a circular dining table. In how many ways can they be arranged if the two hosts are separated? 2

Hosts separated = Total — hosts together.

$= 7! - (1 \times 2 \times 6!) = 3600$

10 At a fund-raising dinner, each rectangular table has 9 seats, 5 facing the stage and 4 with their backs to the stage. In how many ways can nine people be seated at a particular table if:

a. Angelo and Agam must sit on the same side? 2

same side

Both facing stage : $^5P_2 \times 7!$

Both not facing : $^4P_3 \times 7!$

Both on the same side = 161 280

b. Angelo and Agam must sit on opposite sides? 2

on opposite sides :

$= 2\left[\,^5P_1 \times {}^4P_1 \times 7!\,\right] = 201\ 600$

11 A box contains 2 red, 3 blue, 9 green and 9 yellow balls. What is the minimum number of balls that must be drawn to be certain that there will be 4 balls of the same colour? 2

2 R, 3 B, 9 G, 9 Y

Minimum number drawn to guarantee 4 balls of same colour

2 + 3 + 3 + 3 + 1 = 12

12 A PIN is to be chosen using four of the digits 0 through 9. Digits cannot be repeated. How many possible PINs are there? 1

$10 \times 9 \times 8 \times 7 = 5040$ Or ${}^{10}P_4 = 5040$

13 Four Year 12 students, three Year 11 students, and two Year 10 students are seated around a circular table randomly.

a. How many seating arrangements are possible? 1

$8! = 40320$

b. What is the probability that the three groups are seated together? 2

$$\frac{4! \times 3! \times 2! \times 2!}{40320} = \frac{576}{40320} = \frac{1}{70}$$

14 A school committee consists of 8 members and a chairperson. The members are selected from 12 students. The chairperson is selected from 4 teachers. In how many ways could the committee be selected? 1

selection from the group and the order is not important

$${}^{12}C_8 \times {}^{4}C_1$$

15 A sporting team needs 5 members, which must include at least two students from Year 11 and at least two students from Year 12. There are ten Year 11 students and fifteen Year 12 students available for selection. In how many distinct ways can the team be chosen? 1

$${}^{10}C_2 \times {}^{15}C_3 + {}^{10}C_3 \times {}^{15}C_2 = 33\,075$$

16 Lily and Ben are two of eight people.

a. Find the number of ways that the 8 people can arrange themselves in a circle so that Lily and Ben are together. 1

$6! \times 2 = 1440$

b. Find the number of ways that the 8 people can arrange themselves in a straight line so that Lily is somewhere to the left of Ben. 1

$\frac{8!}{2!} = 20160$

16 A bag contains ten tickets numbered 1, 2, 3……,10. Joanna chooses a ticket at random, records the number and **replaces** the ticket in the bag. The process is repeated two more times so that a total of three tickets are drawn. What is the probability that the number on the first ticket drawn is higher than the second AND the second is higher than the third? 2

Total number of outcomes $= 10 \times 10 \times 10 = 1000$

Number of ways of picking 3 distinct numbers $= 10 \times 9 \times 8$

Number of arrangements of each set of 3 distinct numbers $= 6$

$\therefore \frac{1}{6}$ of arrangements will be in descending order

$$\therefore \text{Probability} = \frac{1}{6} \times \frac{10 \times 9 \times 8}{1000} = \frac{3}{25}$$

17 A school has ten prefects. John and Alex are two of those prefects. A committee of six prefects is to be formed to organise a function.

a. How many different committees can be formed? 1

$$\text{Number of different committees} = {}^{10}C_6 = 210$$

b. What is the probability that the committee formed contains John but does not contain Alex? 1

Jun is chosen. 5 other prefects chosen in ${}^{8}C_5$ ways (given Alex not chosen)

$$\text{Prob} = \frac{{}^{8}C_5}{{}^{10}C_6} = \frac{4}{15}$$

18 A bag contains 5 identical blue marbles, 6 identical black marbles and 3 identical red marbles. 2
Three marbles are drawn at random. Find the probability that exactly two blue marbles are drawn. (Answer in simplified fraction form).

5 Blue 6 Black 3 Red

Exactly 2 Blue in 3 = ${}^{5}C_2 \times {}^{9}C_1 = 90$

Arrangements of 3 from 14 = ${}^{14}C_3 = 364$

$\therefore P(E) = \frac{90}{364}$ OR $\frac{45}{182}$